

Exp No:16**SERIAL INTERRUPT USING UART (KEYBOARD INPUT) IN KEIL****AIM:**

To write and simulate an Embedded C program using serial interrupt in 8051 microcontroller such that while a message is continuously transmitted, any key pressed from the keyboard generates an interrupt and is displayed through UART.

APPARATUS / SOFTWARE REQUIRED:

- PC with Windows OS
- Keil μ Vision IDE
- 8051 Simulator (Keil built-in)

- Serial Window (UART #1)

PROGRAM:

```
#include <reg51.h>;
#include <stdio.h>;

void serial_ISR(void) interrupt 4
{
    char ch;
    if (RI)
    {
        RI = 0; // Clear receive interrupt flag
        ch = SBUF; // Read received character

        printf("\n\nINTERRUPT RECEIVED: ");
        printf("%c\n", ch);
    }
}

void delay(void)
{
    unsigned int i;
    for(i = 0; i < 50000; i++); // Delay for visibility
}

void main(void)
{
    SCON = 0x50; // UART mode 1, 8-bit, REN enabled
    TMOD = 0x20; // Timer1 mode 2 (auto-reload)
    TH1 = 0xFD; // Baud rate 9600
    TR1 = 1; // Start Timer1
}
```

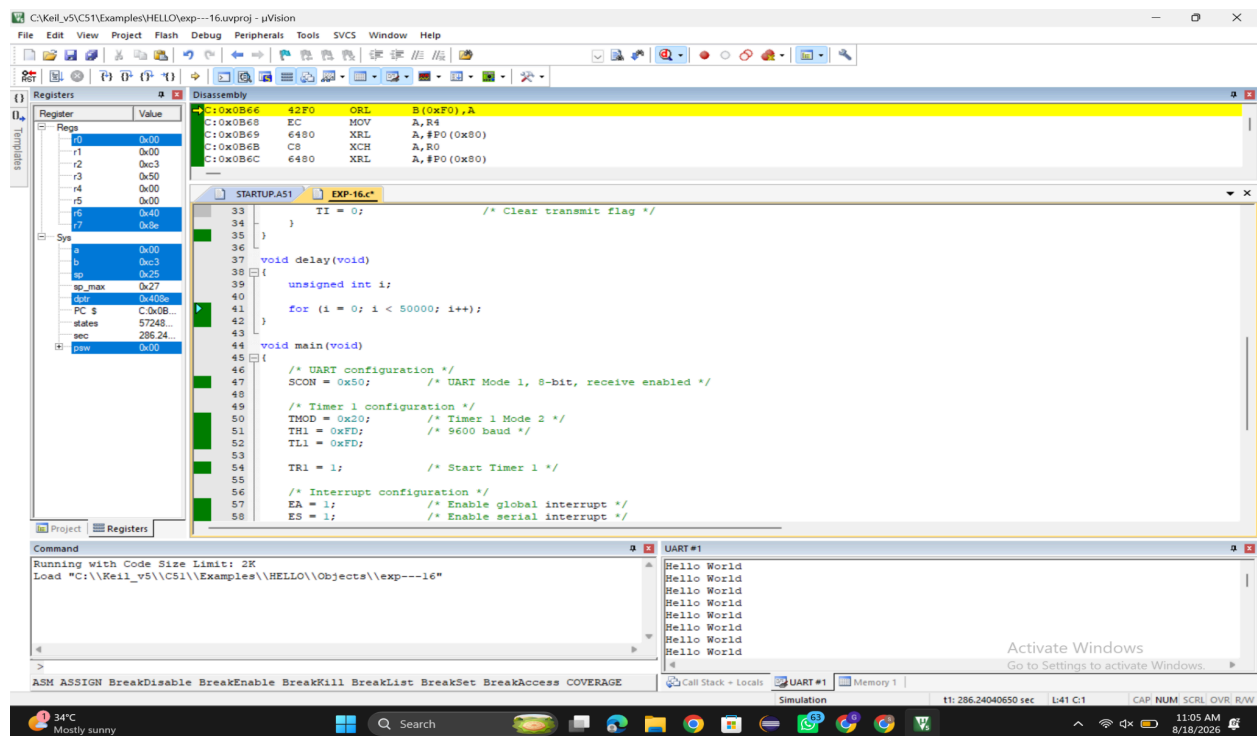
IE = 0x90; // EA = 1, ES = 1 (enable serial interrupt) TI

= 1; // Enable printf

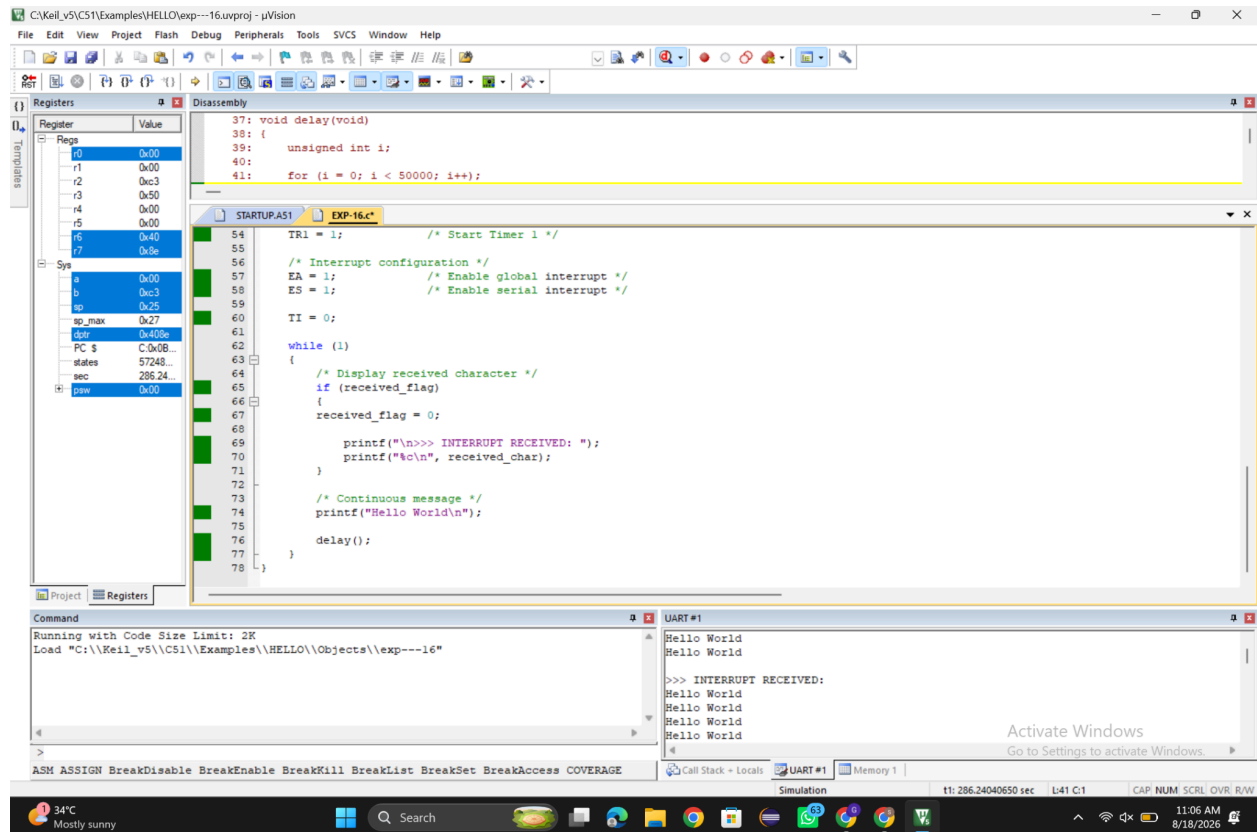
```
while(1)
{
    printf(&quot;Hello World\n&quot;);
    delay();
}
```

OUTPUT:

a) Displaying a Message:



b) During interruption:



RESULT:

Thus, the serial interrupt using UART was successfully implemented and verified in Keil simulator. The interrupt was triggered upon keyboard input and serviced correctly using an Interrupt Service Routine.